Industrial Internship Report on

" Object Counter System"

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was "Object Counter System." The objective of this project was to develop an automatic object counting system using Arduino Uno, two ultrasonic sensors, and an OLED display. The system detects objects entering and leaving a monitored area and updates the object count automatically. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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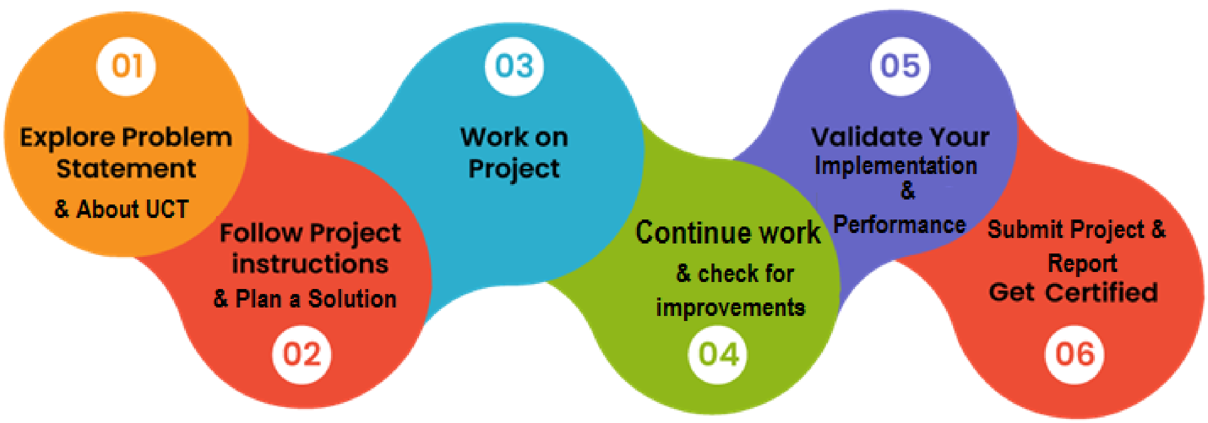
1 Preface

This industrial internship provided me with an opportunity to understand how real-world engineering problems are solved using Embedded Systems and IoT concepts. During the six-week internship, I worked on the **Object Counter System**, which automatically counts objects entering and leaving a monitored area using ultrasonic sensors and displays the count on an OLED display.

The internship helped me understand the complete development process, including problem analysis, solution design, programming, testing, and documentation. It improved my knowledge of Arduino programming, sensor interfacing, OLED display integration, and embedded system development.



The internship was well planned, beginning with understanding the problem statement, followed by designing the solution, implementing the project, testing its performance, and preparing the final report.

I sincerely thank **upskill Campus**, **The IoT Academy**, **UniConverge Technologies Pvt. Ltd. (UCT)**, and my college faculty for providing this valuable learning opportunity and their continuous support.

I encourage my juniors and peers to participate in such internships, as they provide practical exposure, improve technical skills, and help bridge the gap between academic learning and real-world applications.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

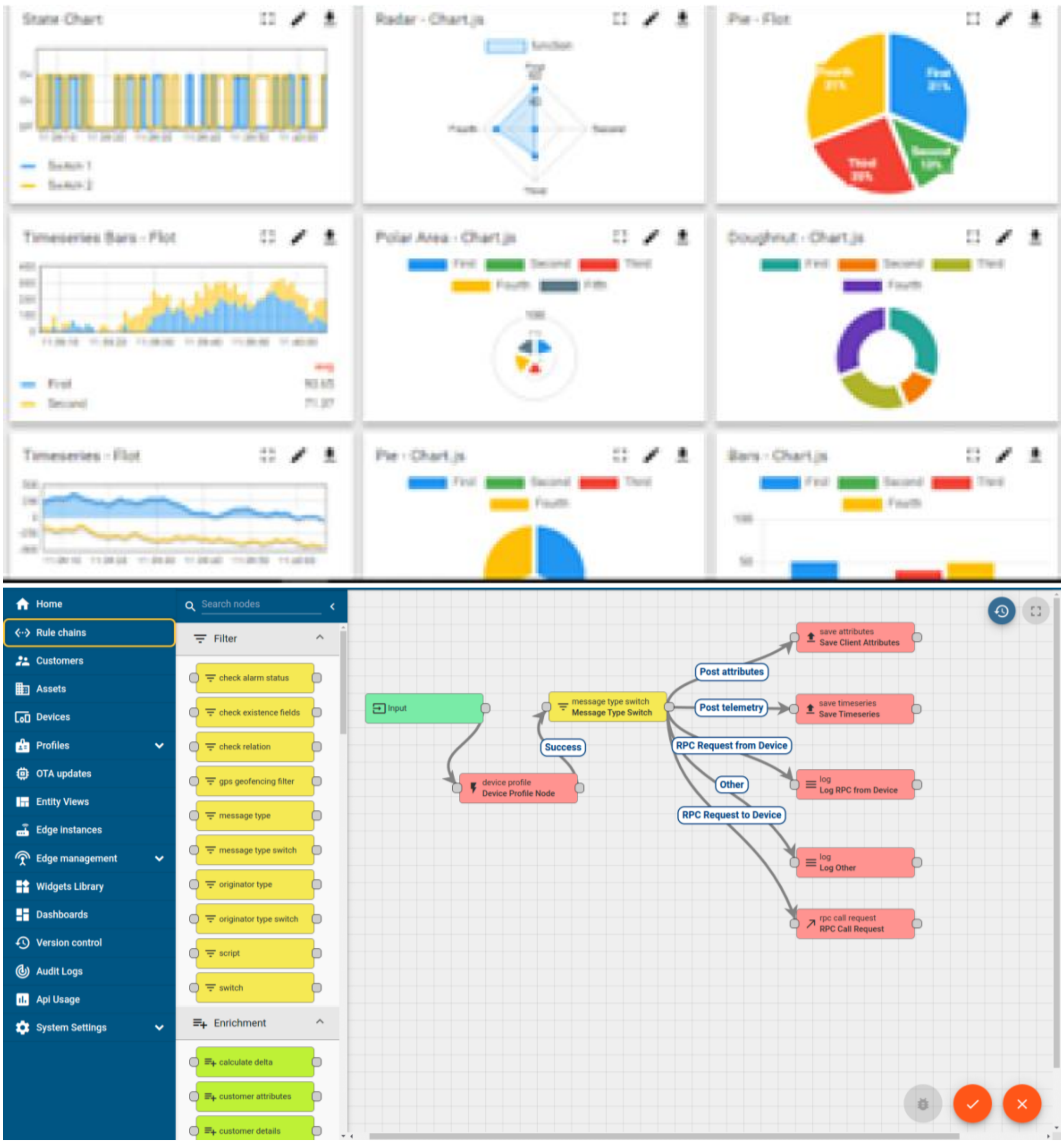
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

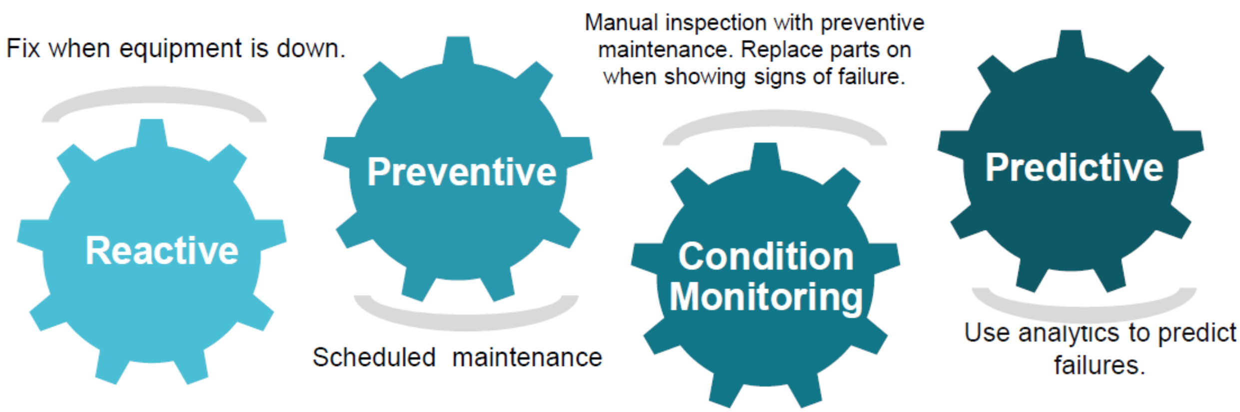
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

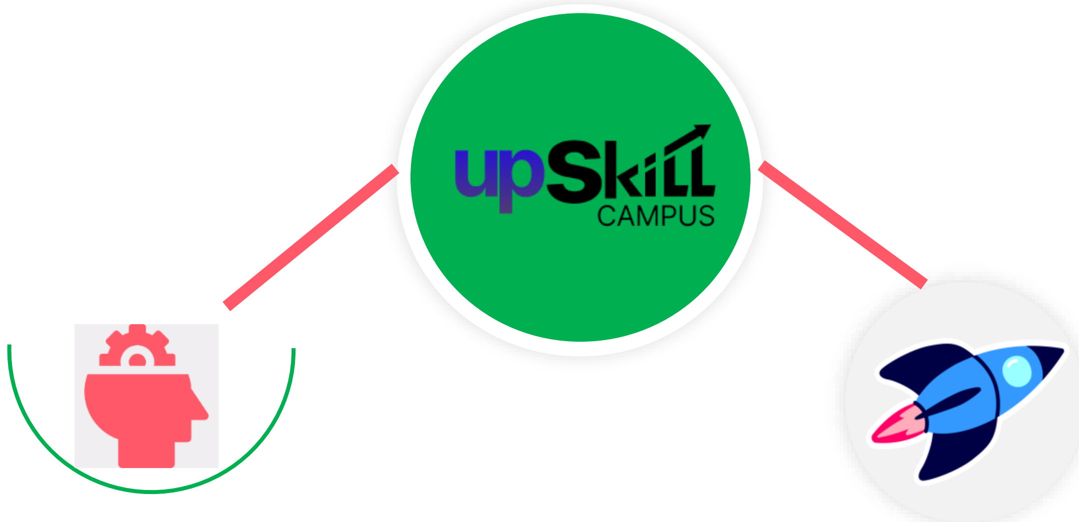
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

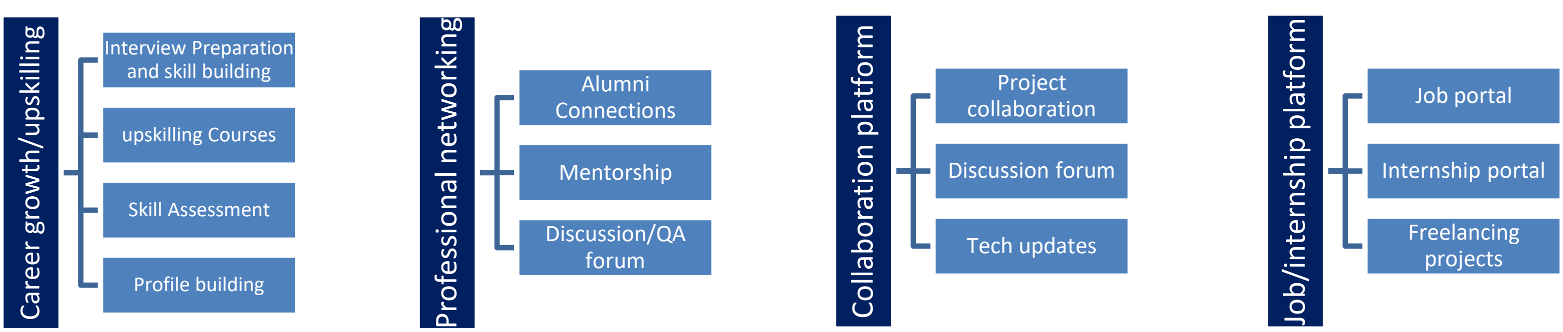
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Arduino Official Documentation – https://www.arduino.cc

[2] Adafruit SSD1306 OLED Library Documentation – https://learn.adafruit.com

[3] HC-SR04 Ultrasonic Sensor Datasheet and Technical Documentation.

2.6 Glossary

Terms	Acronym
Arduino Uno	MCU
Ultrasonic Sensor	HC-SR04
Organic Light Emitting Diode	OLED
Inter-Integrated Circuit	I^2C
Input/Output	I/O

3 Problem Statement

Object counting is an essential requirement in industries, warehouses, railway stations, airports, shopping malls, and other public places where monitoring the movement of objects or people is important. Traditional manual counting methods are time-consuming, inefficient, and prone to human errors, especially in high-traffic environments.

The objective of this project is to develop an **Object Counter System Using Arduino** that automatically counts objects entering and leaving a monitored area. The system uses **two ultrasonic sensors** placed at the entry and exit points to detect the direction of movement. When an object passes the entry sensor followed by the exit sensor, the count increases. Similarly, when an object passes in the opposite direction, the count decreases. The updated count is displayed on an **OLED display** in real time.

This automated solution improves counting accuracy, reduces manual effort, and provides a simple and cost-effective method for object monitoring.

4 Existing and Proposed solution

Existing Solution

Existing object counting systems commonly use **IR sensors**, **camera-based vision systems**, and **laser sensors**. These systems are widely used in industries, warehouses, shopping malls, railway stations, and airports for automatic object counting. Although they are effective, they have certain limitations.

Limitations

- High installation and maintenance cost.
- Camera-based systems require image processing and powerful hardware.
- Laser-based systems are expensive for small-scale applications.
- IR sensor systems may be affected by environmental conditions and sensor alignment.

Proposed Solution

The proposed solution is an **Object Counter System Using Arduino Uno, two ultrasonic sensors, and an OLED display**. The two ultrasonic sensors are placed at the entry and exit points to detect the direction of object movement. Based on the sequence of sensor activation, the Arduino automatically increases or decreases the object count and displays the updated count on the OLED display in real time.

Value Addition

- Cost-effective and easy-to-implement solution.
- Accurate detection of object entry and exit using two HC-SR04 Ultrasonic Sensors.
- Real-time object count displayed on the OLED Display.
- Suitable for industries, warehouses, shopping malls, and other object monitoring applications.

4.1 Code submission (Github link)

https://github.com/Tanuja218/upskillcampus

4.2 Report submission (Github link) :

https://github.com/Tanuja218/upskillcampus

5 Proposed Design/ Model

The proposed **Object Counter System** is designed using **Arduino Uno**, **two HC-SR04 Ultrasonic Sensors**, and an **OLED Display**. The system automatically detects the direction of object movement by monitoring the sequence of sensor activation. Based on the detected movement, the Arduino Uno updates the object count and displays it on the OLED Display in real time. The design is simple, reliable, and suitable for automatic object monitoring applications.

5.1 High Level Diagram

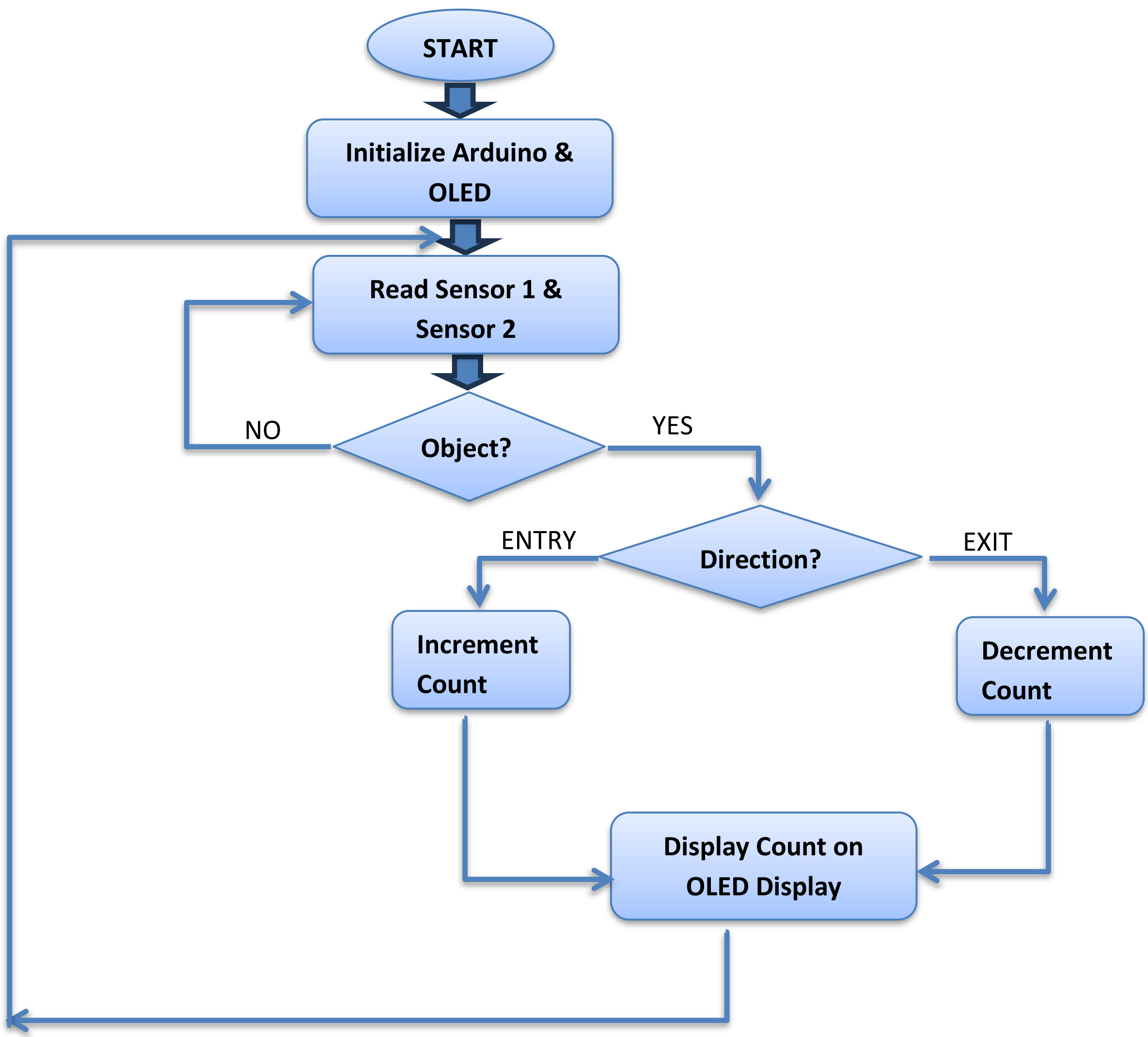


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

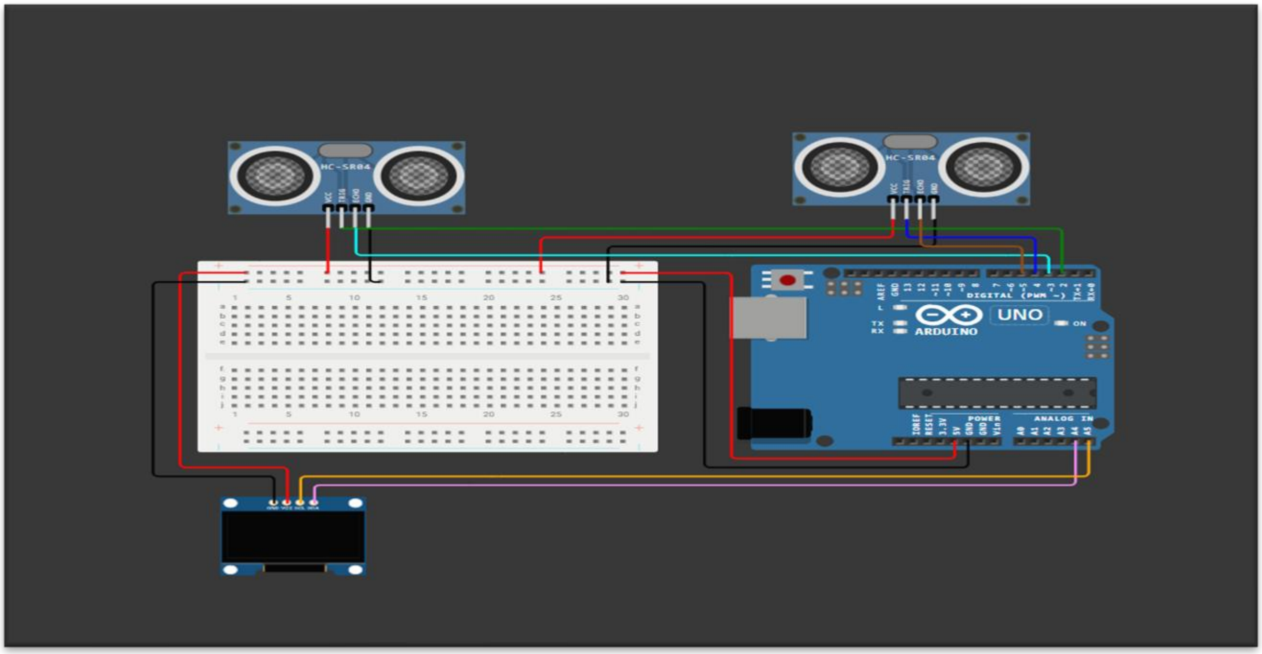
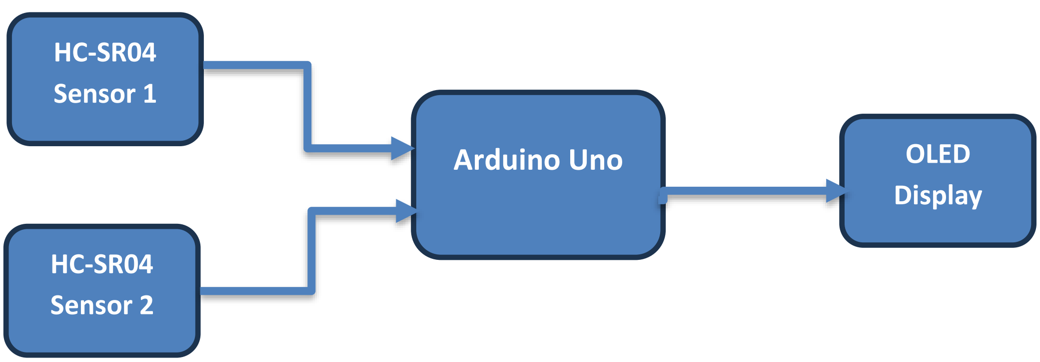


Figure 2 : LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces (if applicable)



The interface consists of two HC-SR04 ultrasonic sensors, an Arduino Uno, and an OLED display. The ultrasonic sensors send distance measurements to the Arduino Uno, which processes the data to determine object movement. The updated object count is transmitted to the OLED display through the **I²C communication protocol** for real-time monitoring.

6 Performance Test

The developed **Object Counter System** was tested to evaluate its accuracy, response, and reliability in detecting object movements. The system was tested under different entry and exit conditions to verify that the object count was updated correctly on the OLED display.

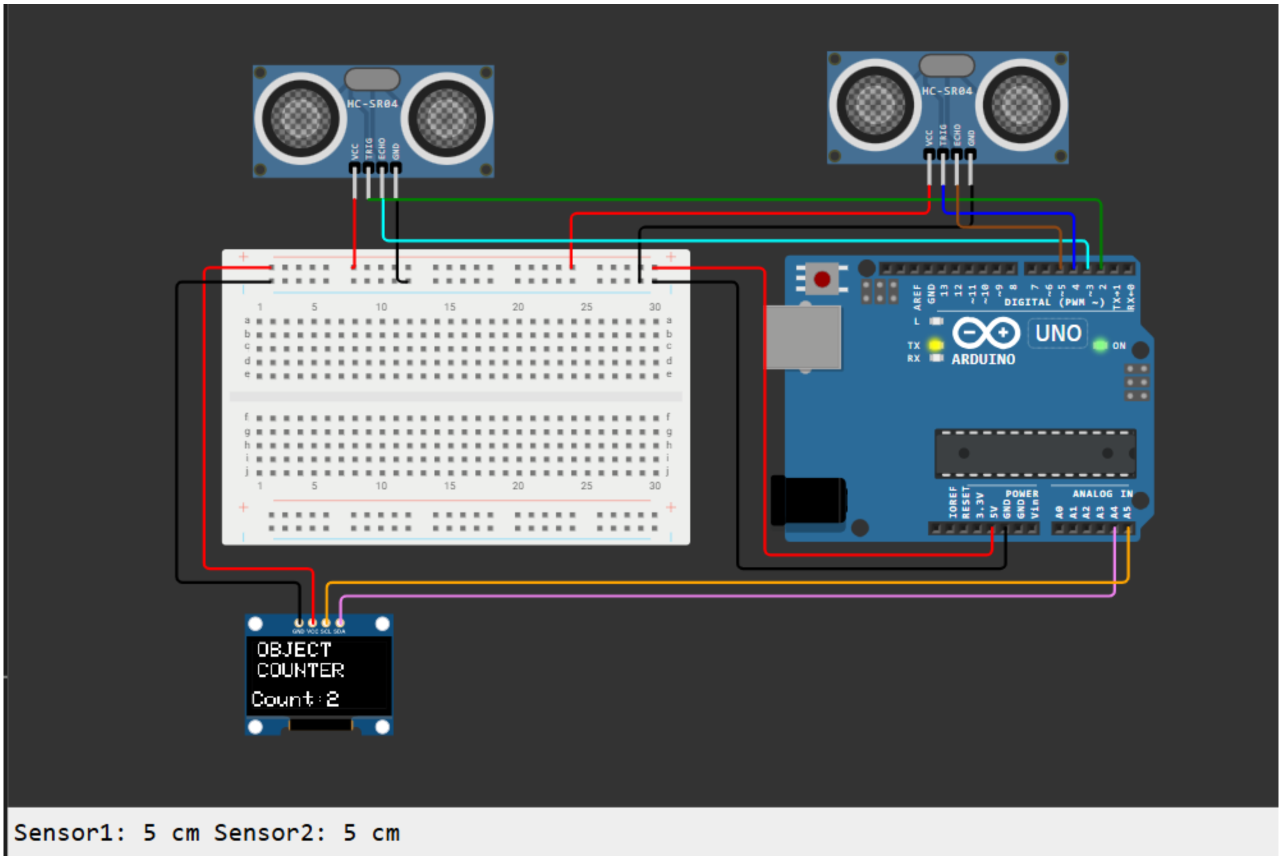
6.1 Test Plan/ Test Cases

Test Case	Expected Result	Status
Detect object at entry	Count increases by 1	Passed
Detect object at exit	Count decreases by 1	Passed
Verify OLED display	Updated count displayed correctly	Passed
Multiple entry and exit movements	Accurate counting without errors	Passed

6.2 Test Procedure

- Powered on the Arduino Uno and initialized the system.
- Moved an object near the entry sensor and verified that the count increased.
- Moved an object near the exit sensor and verified that the count decreased.
- Checked the OLED display after each movement.
- Repeated the tests to verify consistent operation.

6.3 Performance Outcome



The developed **Object Counter System** successfully detected object movements using two **HC-SR04 Ultrasonic Sensors**. The **Arduino Uno** processed the sensor inputs and accurately updated the object count on the **OLED Display** in real time. The system demonstrated reliable performance, fast response, and consistent operation during testing.

7 My Learnings

During this internship, I gained valuable knowledge about **Embedded Systems** and their practical applications. The learning materials provided by **UpSkill Campus** helped me understand the fundamentals of Arduino programming, sensors, microcontrollers, circuit design, and embedded system development. The internship also introduced me to the complete process of developing an embedded project, from understanding the problem statement to designing, testing, and documenting the solution.

As part of the internship, I developed an **Object Counter System using Arduino Uno, two HC-SR04 Ultrasonic Sensors, and an OLED Display**. Through this project, I learned how to interface hardware components, write Arduino programs, simulate circuits using **Wokwi**, and test the system under different conditions. I also improved my skills in preparing flowcharts, block diagrams, circuit diagrams, and technical reports.

In addition to technical knowledge, this internship improved my **problem-solving, analytical thinking, debugging, documentation, and project management skills**. It also taught me the importance of systematic testing and proper project documentation. Overall, this internship provided valuable hands-on experience and increased my confidence in developing embedded system applications. The knowledge and skills gained during this internship will be beneficial for my future academic projects and professional career.

8 Future work scope

The developed **Object Counter System** can be further improved by adding advanced features and technologies. In the future, the system can be integrated with **IoT platforms** to enable remote monitoring and real-time data logging through the internet. A mobile or web application can also be developed to display the object count and system status from any location.

The accuracy of the system can be enhanced by using **camera-based object detection** or **AI-based image processing** for complex environments. Additional features such as **data storage**, **wireless communication (Wi-Fi/Bluetooth)**, **automatic report generation**, and **battery backup** can also be incorporated to improve reliability and usability. These enhancements would make the system suitable for large-scale industrial, commercial, and smart monitoring applications.